



MINING DEALER BEST PRACTICE

Expanded Mining Products In-Pit Workshop

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

Expanded Mining Products In-Pit Workshop.....	1
1.0 Introduction.....	2
2.0 Best Practice Description.....	2
3.0 Implementation Steps	5
4.0 Benefits.....	5
5.0 Resources Required	6
6.0 Supporting Attachments / References	6
7.0 Related Best Practices	6
8.0 Acknowledgements.....	6

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices.” Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

**August 19
1.01-190320-02**

1.0 Introduction

Hydraulic mining shovels, electric rope shovels, draglines and rotary drills work deep inside the pit, which is generally far away from the main workshop. Trimming these machines would prove to be very time consuming and costly. In some cases, additional equipment and labor must be brought in – For example, a water truck would be required to water down the undercarriage to prevent overheating and subsequent failure of track rollers. Therefore, service work on these machines must be performed within the pit in a designated flat area (shovel pad/drill pad)

This Best Practice describes the use of an in-pit workshop that can be placed at the shovel pad or drill pad. Additional considerations are also included to the health & safety of technicians. As the active mining faces progress, this workshop can be moved to a more suitable position.

2.0 Best Practice Description

Instead of working under the weather or sheltering in an empty container, an in-pit workshop can be built and placed at the shovel pad to execute service work faster and more efficiently. The in-pit workshop can either be self-built locally, bought as a standard unit, or built by a supplier to specifications.

The pictures in this best practice show one unit built by a local supplier in Australia. The pictures are reproduced with permission from the supplier and Caterpillar does not endorse this specific supplier. It is simply shown as an example of the possibilities.

Figure 1 shows a standard ready-made unit with side opening doors – Converted from a standard container, this option allows for easy transportation. The doors can be locked in the open position. When opened, they act as side walls on both sides and have fold-down work benches. A roof is placed at the top to shed the container from a harsh sun for better cooling. Extendable awnings can be wound out to extend the shaded area. A lockable store room is built in for storage of tools and parts.

Figure 2 shows a separate unit which is a combined wash room and lunch room. Separate units can be introduced for higher occupancy rates. Access ramps provide safe entry and exit to both areas. They can be built as foldable units or removal and storage in the lunch room for transportation. The air conditioning unit is flushed with the external side wall to avoid damage during transportation.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Expanded Mining Products In-Pit Workshop	DATE 8/1/2019	CHG NO 00	NUMBER 1.01-190320-02

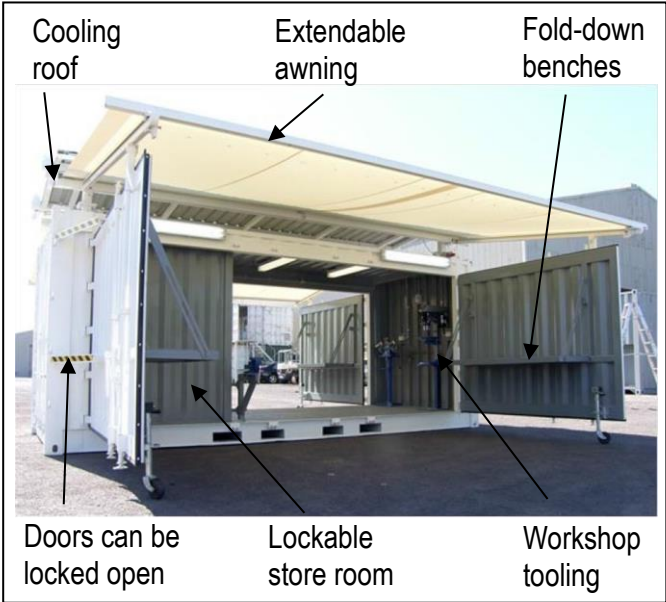


Fig 1: New mobile workshop from brochure.
(Port Container Services Pty Ltd)

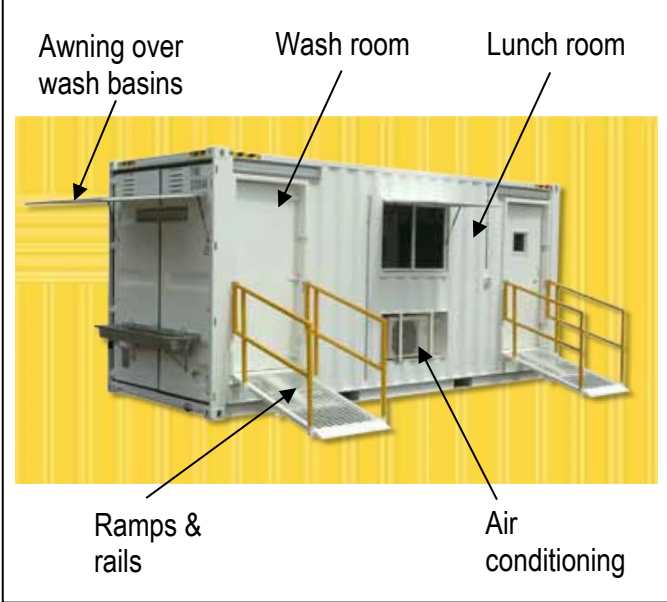


Fig 2: Wash room and lunch room as one unit.
(Port Container Services Pty Ltd)

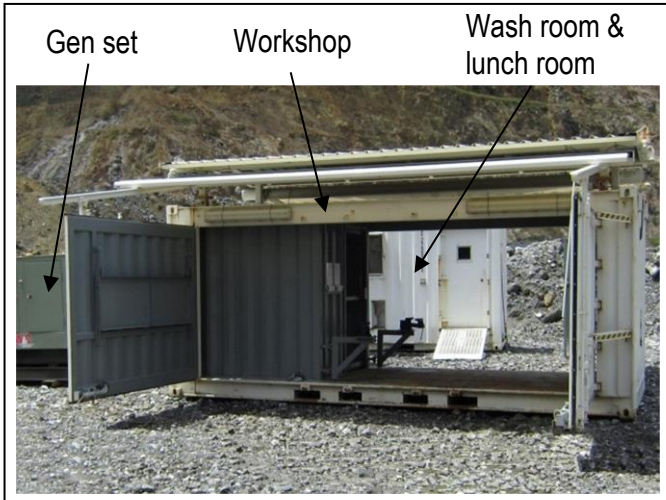


Fig 3: Set up at one mine site



Fig 4: Store room for parts and tools.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Expanded Mining Products In-Pit Workshop	DATE	CHG NO	NUMBER
	8/1/2019	00	1.01-190320-02

Figure 3 is an existing setup on site. A generator set is also included in the setup to provide power for lighting, electrical tools, and utilities in the lunch/wash room (The lunch room module is set on the 'back side' of the workshop module.) The castor wheels attached to the doors are shown in their stowed position.

Figure 5 shows a general layout plan at a shovel pad. The usual safety perimeter and barriers need to be set up, with the ground leveled and electrical cables set out of the way to prevent trips and falls. The gen set should also be placed in a relation with prevailing winds blowing fumes away from the modules.

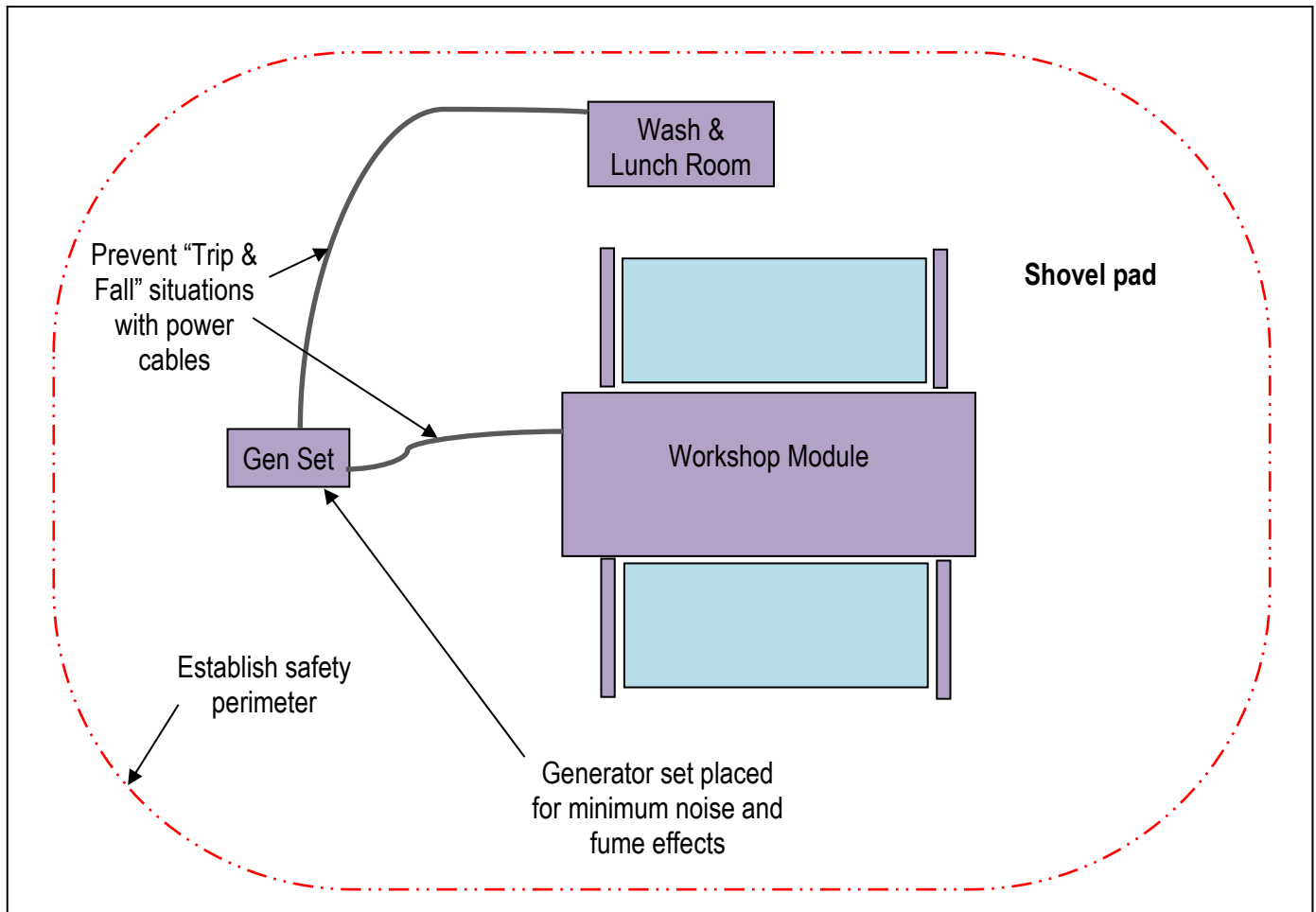


Fig 5: General layout plan.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Expanded Mining Products In-Pit Workshop	DATE	CHG NO	NUMBER
	8/1/2019	00	1.01-190320-02

3.0 Implementation Steps

To implement such a project, a champion needs to be elected to research and manage the process of introducing such a setup.

- i. Research on suitable suppliers and products available.
- ii. Obtain budgetary pricing with contingency costs for justification.
- iii. Develop justification costing model.
- iv. Research on requirements by technicians.
- v. Determine power requirements for suitable gen set.
- vi. Research on site requirements and legislations.
- vii. Review suitability and availability of equipment for transporting modules.
- viii. Select suitable supplier.
- ix. Develop and finalize on requirements of the modules with supplier.
- x. Obtain authorization to purchase.
- xi. Develop instructions on usage of the workshop.
- xii. Develop instructions for starting and shutting down of gen set.
- xiii. When delivered, commission and conduct training on usage of the whole setup.
- xiv. Move to desired location.

4.0 Benefits

There are a few different benefits in using such a setup.

- i. Savings in technician labor due to availability of common parts being stored in the mobile workshop.
- ii. Savings in technician labor with less frequent traveling to and from the main workshop when work involves component repairs.
- iii. Improved work quality that can impact reliability and improve MTBS (Mean Time Between Shutdowns).
- iv. Lower downtime so that machine can get back to work earlier for higher production.

As an example, assume:

A shovel moving 3,000 tonnes of material per hour in a gold mine with a 1:4 strip ratio.

Ore has 10gm gold per tonne and gold recovery is 85%.

Utilization is 70%.

400 repair jobs per year, 40% of which can make use of the in-pit workshop to an advantage.

Savings of 15 minutes for each of these jobs.

Additional hours available = $400 \times 40\% \times (15/60) \times 70\% = 28$ hours per year

Additional ore moved = $(3000 \div 5) \times 28 = 16,800$ tonnes per year

Additional gold production = $16,800 \times (10 \div 31) \times 85\% = 4,606$ oz gold per year

Additional benefit @ \$1200 per ounce = $4,606 \times 1200 = \$5.5M$ per year

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Expanded Mining Products In-Pit Workshop	DATE 8/1/2019	CHG NO 00	NUMBER 1.01-190320-02

5.0 Resources Required

The project of introducing this setup on site requires a Champion. The Champion is required to source and cost the project. The benefits can then be presented to management for approval. When approved, the Champion will oversee the development of detailed specifications, build-up, purchase, document development and commissioning of the setup. In most instances, the site should already have a shovel or drill pad in the vicinity of the load area. If this is not the case, the Champion will have to work with Production and Mine Engineers on location of such an area.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

Idea & pictures:

Luke Seaton

Product Support Representative

Hastings Deering Australia Limited

Queensland, Australia

Luke.Seaton@hastingsdeering.com.au

Permission to utilize supplier pictures/information:

Luke Stevens

Container Technician

Port Container Services

Tel: +614 1307 7899

luke@port.com.au

www.portcontainerservices.com.au

Written by:

Victor Lai

Equipment Management Consultant

Caterpillar Global Mining

lai_victor_k@cat.com

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Expanded Mining Products In-Pit Workshop	DATE 8/1/2019	CHG NO 00	NUMBER 1.01-190320-02